

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter End Semester Examination – Dec. 2019

Course: Electrical Engineering

Sem: IV

Subject: Introduction to Nonconventional Energy Sources

Subject Code: BTEEOE 407B

Max Marks: 60

Date: 06/12/2019

Duration: 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
 2. Attempt any five questions of the following.
 3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
 4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.
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Q.1

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|---|---|
| A) Briefly Explain power scenario in India. | 6 |
| B) Explain the solar radiation Geometry. | 6 |

Q.2

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|--|---|
| A) Explain the Photovoltaic cell with its current voltage characteristics. | 6 |
| B) What is solar pond? Explain working of solar pond. | 6 |

Q.3

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| A) Draw neat labelled diagram of wind turbine and Explain its Components. | 6 |
| B) How much power is contained in wind which blows from one point to another point? Derive its equation. | 6 |

Q.4

- A) How geothermal energy is combined with conventional thermal power plant? Explain in brief with neat diagram. 6**
- B) Enlist different types Geothermal resources and explain with neat diagram Binary steam geothermal power plant. 6**

Q.5

- A) Write a short note on Nuclear fission and Nuclear fusion. 6**
- B) Explain Pressurized heavy water reactor in Nuclear power plant. 6**

Q.6

- A) Explain different technologies to produce biogas? With neat diagram explain Deenbandhu biogas plant. 6**
- B) Explain the production process of ethanol from biogas 6**

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